The Foundations: Logic and Proofs

Chapter 1, Part II: Predicate Logic

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Summary

Predicate Logic

- The Language of Quantifiers
- Logical Equivalences
- Nested Quantifiers
- Translation from Predicate Logic to English
- Translation from English to Predicate Logic

Predicates and Quantifiers

Section 1.4

Section Summary 1

Predicates

- Variables

Quantifiers

- Universal Quantifier
- Existential Quantifier

Negating Quantifiers

- De Morgan's Laws for Quantifiers

Translating English to Logic

Logic Programming (*optional*)

1.4.1 Introduction
Propositional Logic Not Enough

If we have:

"All men are mortal."

"Socrates is a man."

Does it follow that "Socrates is mortal?"

Can't be represented in propositional logic.

Need a language that talks about objects, their properties, and their relations.

1.4.2 Predicates

A Predicate is a declarative sentence whose true/false value depends on one or more variables.

VS

A proposition is a declarative sentence (that is, a sentence that declares a fact) that is either true or false, but not both.

Predicates and Propositional Functions

The statement "x is greater than 3" has two parts:

- the subject: x is the subject of the statement
- the predicate: "is greater than 3" (a property that the subject can have).
- We denote the statement "x is greater than 3" by P(x), where P is the predicate "is greater than 3" and x is the variable.
- The statement P(x) is also called the value of propositional function P at x.

Examples of Propositional Functions$_1$

P(x): “x is greater than 3 ”

Variables x can be replaced by elements from their *domain (U, universe)*.

Assign a value to x, so P(x) becomes a proposition and has a truth value.

P(5)	P(2)
true	false

Examples of Propositional Functions$_2$

Let "$x + y = z$" be denoted by $R(x, y, z)$ and U (for all three variables) be the integers. Find these truth values:

	R(2,-1,5)	R(3,4,7)	R(x, 3, z)
Solution:	F	T	Not a Proposition

Now let "$x - y = z$" be denoted by $Q(x, y, z)$, with U as the integers. Find these truth values:

	Q(2,-1,3)	Q(3,4,7)	Q(x, 3, z)
Solution:	T	F	Not a Proposition

Examples of Propositional Functions$_3$

Given each propositional function determine its true/false value when variables are set as below.

- Prime(x) = "x is a prime number. "

  Prime(2) T Prime(9) F

- C(x, y)="x is the capital of y".

  C(Beijing, China) T C(Hefei,China) F

- E(x , y , z) = "x + y = z".

  E(2, 3, 5) T E(4, 4, 17) F

Predicates and Propositional Functions

- In general, a statement involving the n variables $x_1, x_2, \ldots, x_n$ can be denoted by $P(x_1, x_2, \ldots, x_n)$.
- A statement of the form $P(x_1, x_2, \ldots, x_n)$ is the value of the propositional function P at the *n*-tuple $(x_1, x_2, \ldots, x_n)$,
- and P is also called an *n*-place predicate or an n-ary predicate.

Compound Expressions

Connectives from propositional logic carry over to predicate logic.

If *P(x)* denotes "$x > 0$," find these truth values:

- $P(3) \vee P(-1)$ T
- $P(3) \wedge P(-1)$ F
- $P(3) \rightarrow P(-1)$ F
- $P(3) \rightarrow \neg P(-1)$ T
- $P(3) \wedge P(y)$
- $P(x) \rightarrow P(y)$

Expressions with variables are not propositions and therefore do not have truth values.

When used with quantifiers (to be introduced next), these expressions (propositional functions) become propositions.

1.4.3 Quantifiers

Charles Peirce (1839-1914)

We need *quantifiers* to express the meaning of English words including *all, some, many, none, and few* :

- "We are all Chinese."
- "Some dogs are white."

The two most important quantifiers are:

- *Universal Quantifier,* "For all," symbol: $\forall$
- *Existential Quantifier*, "There exists," symbol: $\exists$

We write as in $\forall x\, P(x)$ and $\exists x\, P(x)$.

$\forall x\, P(x)$ asserts $P(x)$ is true for <u>every</u> x in the *domain*.

$\exists x\, P(x)$ asserts $P(x)$ is true for <u>some</u> x in the *domain*.

Universal Quantifier

$\forall x\ P(x)$ is read as "For all x, P(x)" or "For every x, P(x)"

Examples:

- If $P(x)$ denotes "$x > 0$" and U is the integers, then $\forall x\ P(x)$ is false.
- If $P(x)$ denotes "$x > 0$" and U is the positive integers, then $\forall x\ P(x)$ is true.
- If $P(x)$ denotes "x is even" and U is the integers, then $\forall x\ P(x)$ is false.

Existential Quantifier

$\exists x\ P(x)$ is read as "For some x, P(x)", or as "There is an x such that P(x)," or "For at least one x, P(x)."

Examples:

- If $P(x)$ denotes "$x > 0$" and U is the integers, then $\exists x\ P(x)$ is true. It is also true if U is the positive integers.
- If $P(x)$ denotes "$x < 0$" and U is the positive integers, then $\exists x\ P(x)$ is false.
- If $P(x)$ denotes "x is even" and U is the integers, then $\exists x\ P(x)$ is true.